

9 (**long**) A probe is on a comet, made of porous ice, which is modelled as a sphere of radius R with uniform density ρ . The small probe drills a narrow hole of diameter a through the centre all the way across the comet as shown in Fig. 6.

(a) Ignoring the gravitational pull of the probe, derive an expression for the force on a particle of mass m at radius r in the bore hole. You may assume that the net force on the particle due to all material outside the sphere of radius r represented by the dashed line in the figure is zero, and that the amount of missing mass from the bore hole is negligible compared with the mass of the comet. [8]

(b) Obtain an expression for the work done in lifting to the surface a particle of mass m initially at a distance r from the centre of the comet, in terms of R , ρ , m , r and G (the universal gravitational constant). [6]

(c) Hence derive an expression for the total work done against gravity in lifting out all the material from the bore hole. (Assume that the work done by the drill to break down the material is negligible.) [8]

(d) Derive an expression for the frequency of oscillation of a particle that is released into the empty bore hole. [8]

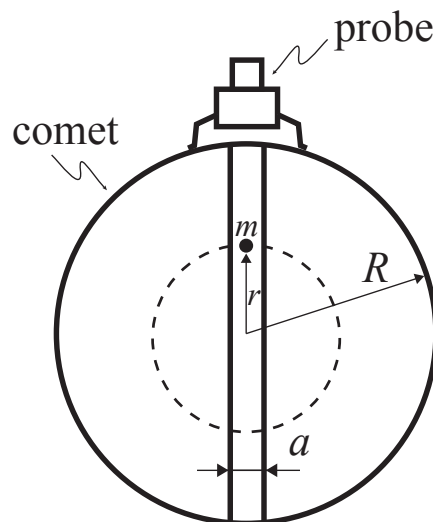


Fig. 6